Network load balancer.

A **Network Load Balancer (NLB)** in AWS is a **high-performance Layer 4 (Transport Layer) load balancer** designed to handle **extreme traffic with ultra-low latency**.

What it actually does .

An NLB **forwards incoming TCP/UDP traffic** to multiple backend targets (EC2, containers, IPs) based purely on **IP + Port** — not content.

I have created three instances with web servers and changed the default port number.

Available instances (1/4)

Filter instances

Instance ID	Name	State	Security groups	Zone
i-087f43c82762f2bb1	port-18	Running	launch-wizard-14	ap-south-1a
i-0dde696fd825fa295	port-25	Running	launch-wizard-14	ap-south-1a
i-0c76205d44ba0096a	port-09	Running	launch-wizard-14	ap-south-1a
i-0bc9a361db492dcec	port-21	Running	launch-wizard-13	ap-south-1a

1st EC2 – 21

2nd EC2 – 25

3rd EC2 – 18

4th EC2 - 09

Then we have to create an target group and assign to the instances.

For EC2 – 1(port-21)

Lambda function
Supports load balancing to a single Lambda function. ALB required as traffic source.
Suitable for: ALB

Application Load Balancer
Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.
Suitable for: NLB

Target group name
Name must be unique per Region per AWS account.

port-21

Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 7/32

Protocol
Protocol for communication between the load balancer and targets.

TCP

Port
Port number where targets receive traffic. Can be overridden for individual targets during registration.

21

1-65535

IP address type
Only targets with the indicated IP address type can be registered to this target group.

IPv4
Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

IPv6
Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). Learn more

VPC
Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

Available instances (1/4)

Filter instances

Instance ID	Name	State	Security groups	Zone
i-087f43c82762f2bb1	port-18	Running	launch-wizard-14	ap-south-1a
i-0dde696fd825fa295	port-25	Running	launch-wizard-14	ap-south-1a
i-0c76205d44ba0096a	port-09	Running	launch-wizard-14	ap-south-1a
i-0bc9a361db492dcec	port-21	Running	launch-wizard-13	ap-south-1a

1 selected

Ports for the selected instances
Ports for routing traffic to the selected instances.

21

1-65535 (separate multiple ports with commas)

Include as pending below

Review targets

Targets (1)

Remove all pending

Filter targets

Show only pending

Instance ID	Name	Port	State	Security groups	Zone	Private IPv4 address	Subnet ID
i-0bc9a361db492dcec	port-21	21	Running	launch-wizard-13	ap-south-1a	172.31.34.92	subnet-032ed1cdaf2ac111d

1 pending

Cancel Previous Next

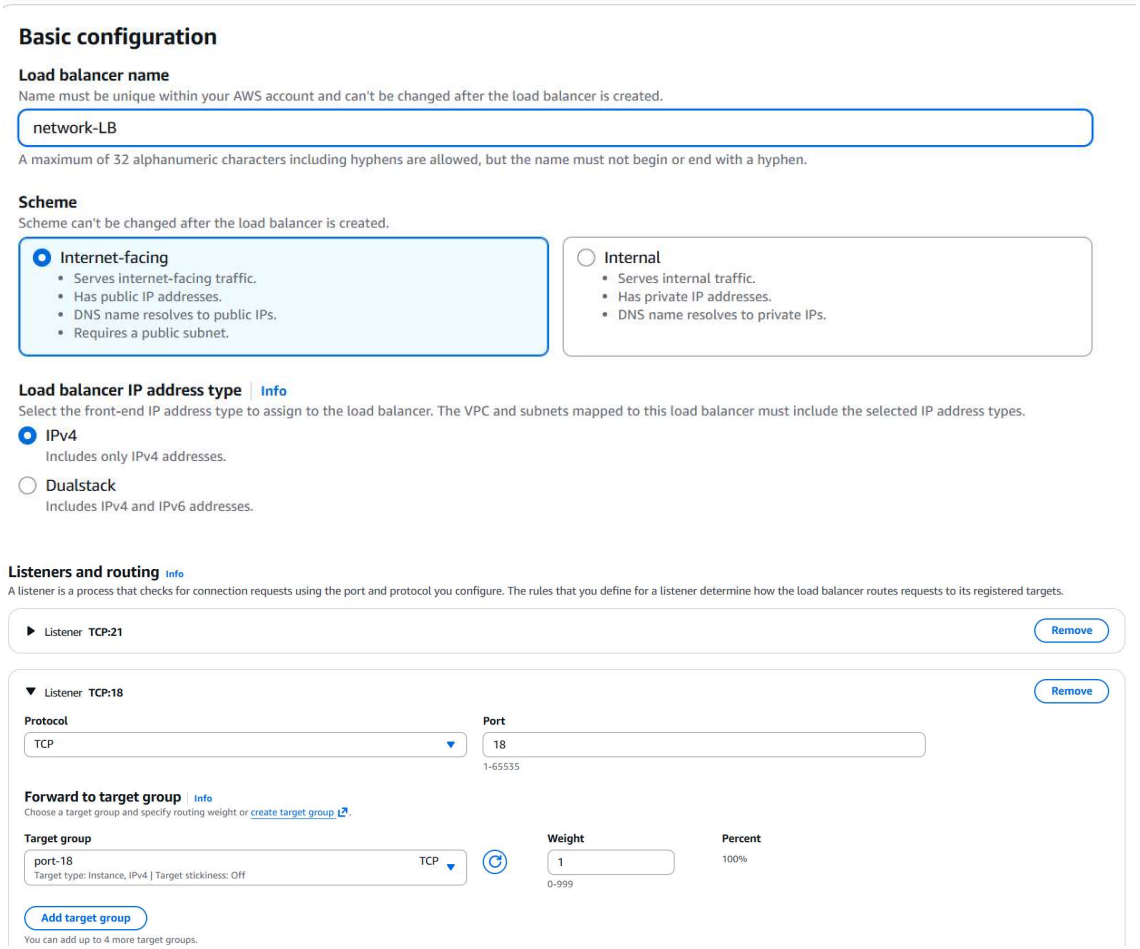
Same procedure for remaining servers

For EC2 – 2

For EC2 – 3

For EC2 – 4

I created the network load balancer for 4 listeners



Then I have selected all the 4 Target groups in the Load balancer.

Listeners (4)

Actions | Add listener

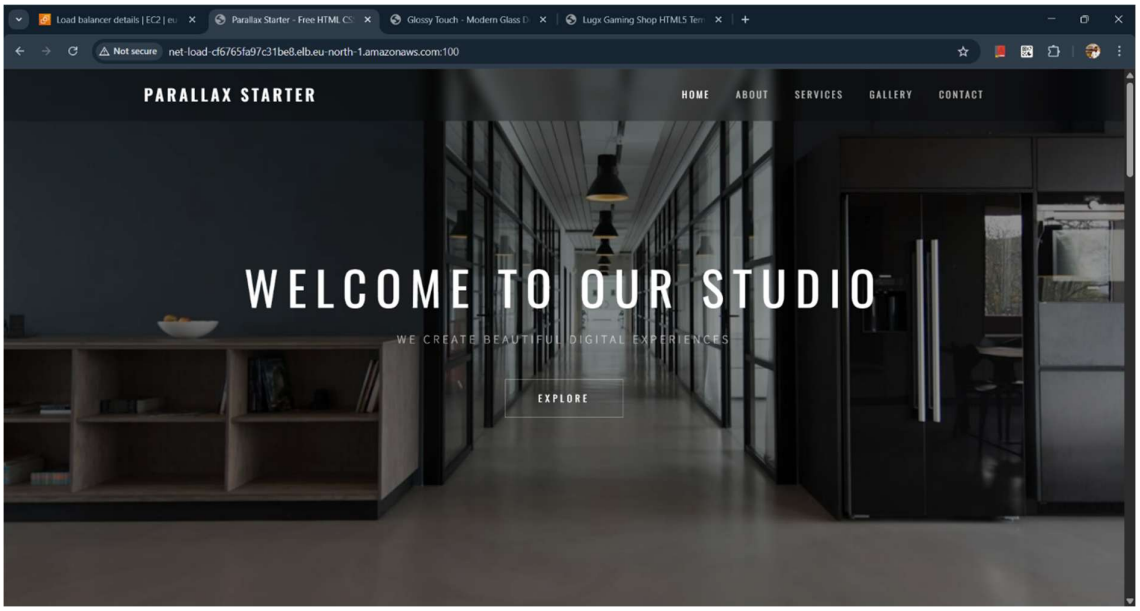
A listener checks for connection requests using the protocol and port that you configure. Traffic received by a Network Load Balancer listener is forwarded to the selected target group.

Filter listeners

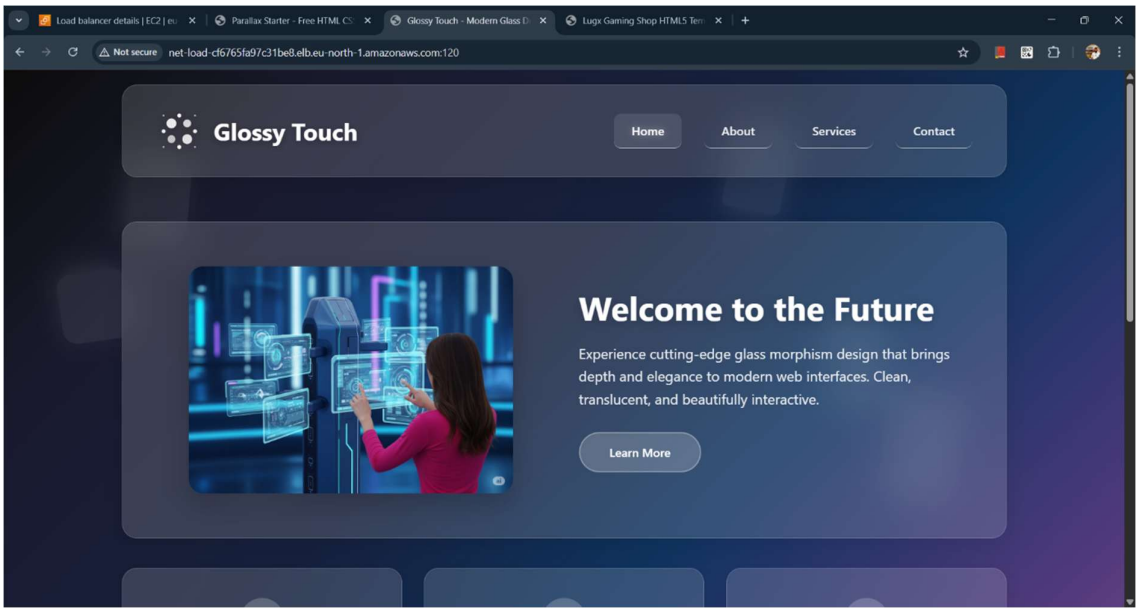
Protocol:Port	Default action	ARN	Security policy	Default SSL/TLS certificate	ALPN policy
TCP:25	**Forward to target group** port-25: 1 (100%) Target group stickiness: Off	ARN	Not applicable	Not applicable	Not applicable
TCP:21	**Forward to target group** port-21: 1 (100%) Target group stickiness: Off	ARN	Not applicable	Not applicable	Not applicable
TCP:9	**Forward to target group** port-09: 1 (100%) Target group stickiness: Off	ARN	Not applicable	Not applicable	Not applicable
TCP:18	**Forward to target group** port-18: 1 (100%) Target group stickiness: Off	ARN	Not applicable	Not applicable	Not applicable

Created Network Load Balancer.

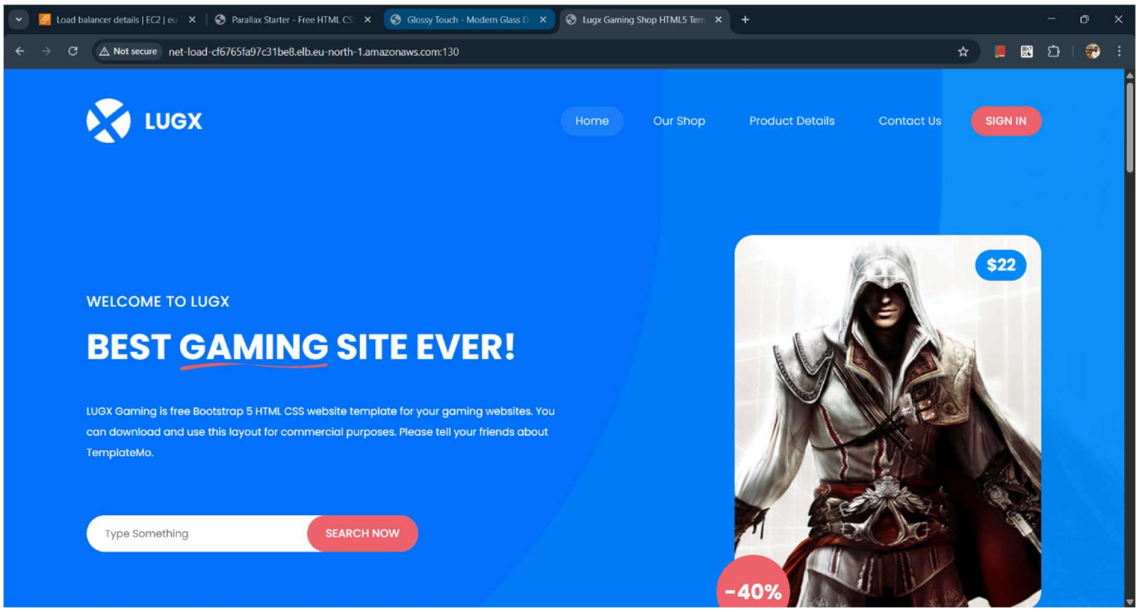
Hosted in Port no 21 (EC2 – 1).



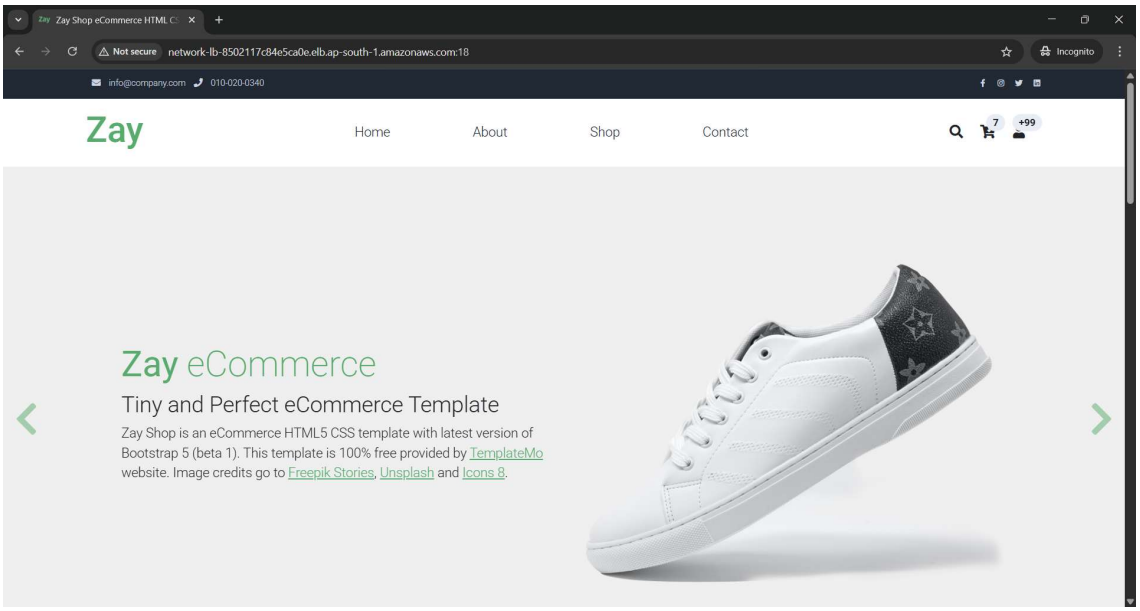
Hosted in Port no 25 (EC2 – 2).



Hosted in Port no 09 (EC2 – 3).



Hosted in Port no 18 (EC2 – 4).


Zay Shop eCommerce HTML C
Not secure
network-lb-8502117c84e5ca0e.elb.ap-south-1.amazonaws.com:18
Incognito
info@company.com
010-020-0340
Zay
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About
Shop
Contact
7
+99
Zay eCommerce
Tiny and Perfect eCommerce Template
Zay Shop is an eCommerce HTML5 CSS template with latest version of Bootstrap 5 (beta 1). This template is 100% free provided by TemplateMo website. Image credits go to Freepik Stories, Unsplash and Icons 8.